CHAPTER

4 CORRELATION ANALYSIS





Karl Pearson

Karl Pearson (1857-1936) was a English Mathematician and Biostatistician. He founded the world's first university statistics department at University College, London in 1911. The linear correlation coefficient is also called Pearson product moment correlation coefficent. It was developed by Karl Pearson with a related idea by Francis Galton (see Regression analysis - for Galton's contribution). It is the first of the correlation measures developed and commonly used.



Charles Spearman

Charles Edward Spearman (1863-1945) was an English psychologist and ,after serving 15 years in Army he joined to study PhD in Experimental Psychology and obtained his degree in 1906. Spearman was strongly influenced by the work of Galton and developed rank correlation in 1904.He also pioneered factor analysis in statistics.

"When the relationship is of a quantitative nature, the appropriate statistical tool for discovering the existence of relation and measuring the intensity of relationship is known as correlation"

—CROXTON AND COWDEN

LEARNING OBJECTIVES

DNBR97

The student will be able to

- learn the meaning, definition and the uses of correlation.
- identify the types of correlation.
- understand correlation coefficient for different types of measurement scales.
- differentiate different types of correlation using scatter diagram.
- calculate Karl Pearson's coefficient of correlation, Spearman's rank correlation coefficient and Yule's coefficient of association.
- interpret the given data with the help of coefficient of correlation.

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